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Aggregate Demand and Aggregate Supply

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Economic Fluctuations

- Now we look at short run fluctuations in economic variables: GDP, Inflation, price level, unemployment, interest rate.

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Economic Fluctuations

- Key facts about economic fluctuations
  1. Economic fluctuations are irregular and unpredictable (unlike what ‘cycle’ implies)
  2. Most macroeconomic variables fluctuate together: Recessions – economy-wide phenomena
  3. As output falls, income falls and unemployment rises

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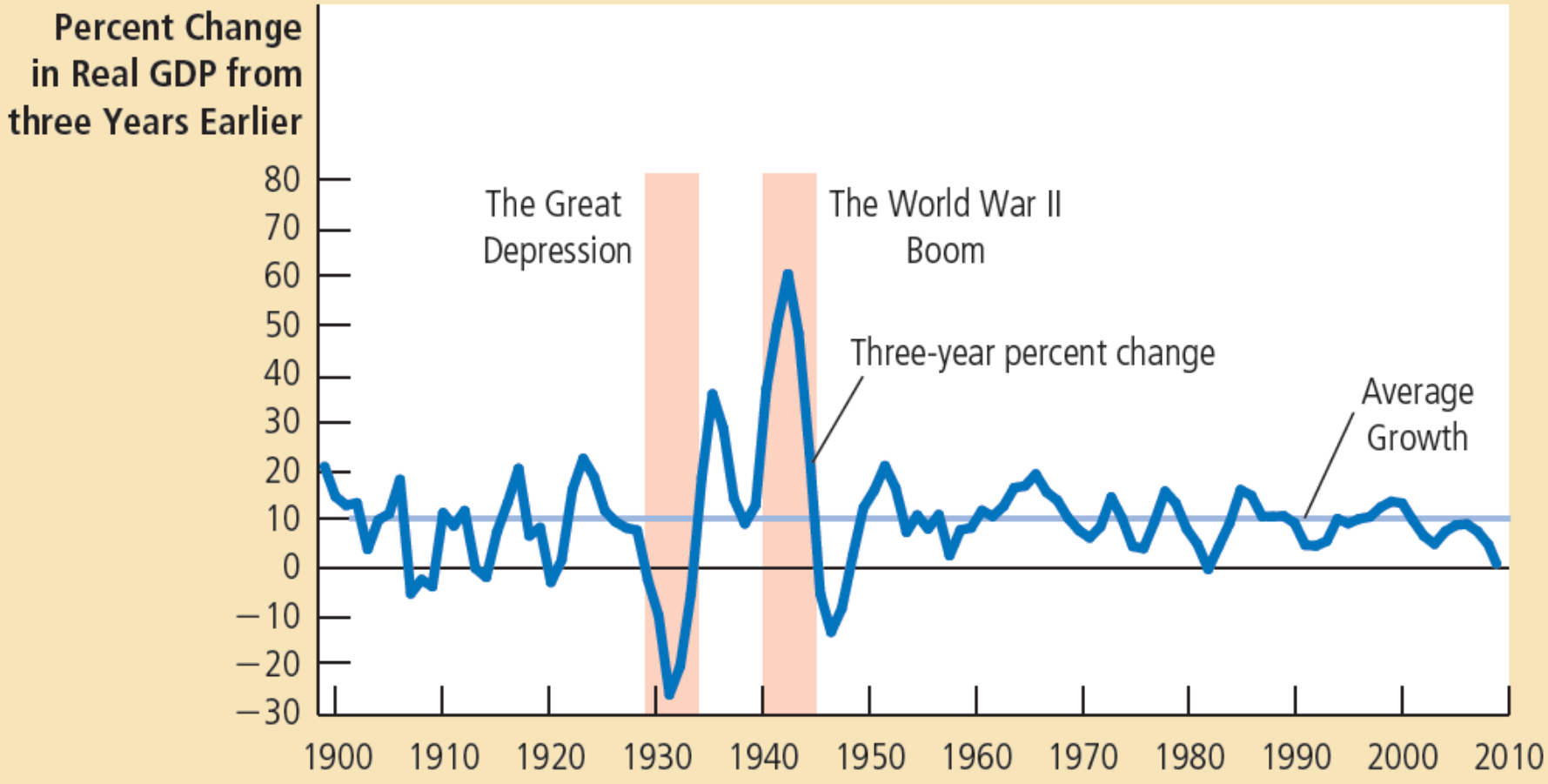
Figure 9

U.S. Real GDP Growth since 1900



Over the course of U.S. economic history, two fluctuations stand out as especially large. During the early 1930s, the economy went through the Great Depression, when the production of goods and services plummeted. During the early 1940s, the United States entered World War II, and the economy experienced rapidly rising production. Both of these events are usually explained by large shifts in aggregate demand.

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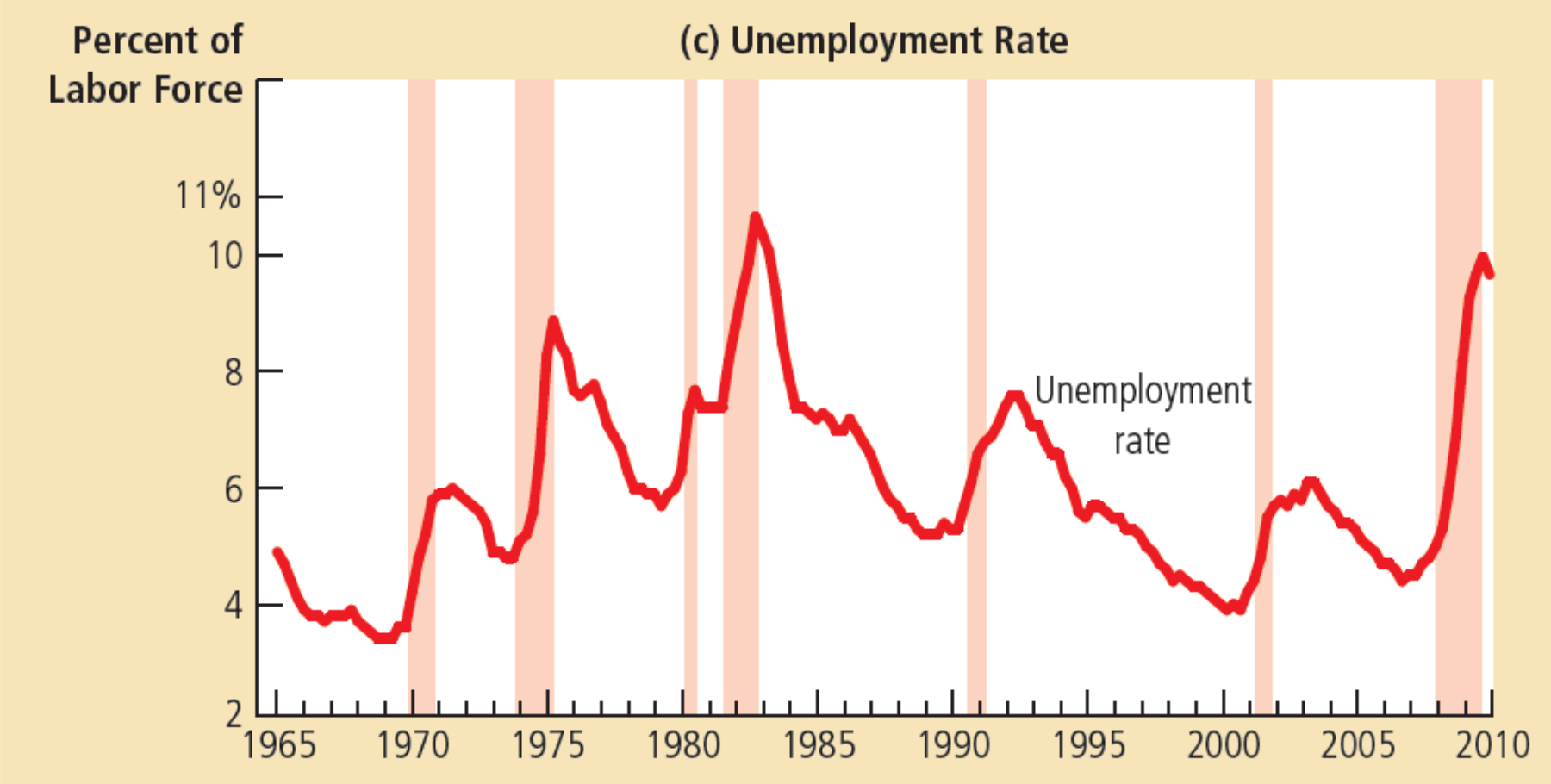
Figure 1

A Look at Short-Run Economic Fluctuations (c)



This figure shows real GDP in panel (a), investment spending in panel (b), and unemployment in panel (c) for the U.S. economy using quarterly data since 1965. Recessions are shown as the shaded areas. Notice that real GDP and investment spending decline during recessions, while unemployment rises.

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Classical Economics

- Long run assumptions: classical dichotomy and monetary neutrality hold
- Thus changes in nominal variables do not affect real variables
- But these assumptions do not hold in short run (year-to-year changes)

Short-Run Economic Fluctuations

- AD-AS model
  - Model of aggregate demand (AD) & aggregate supply (AS)
  - Is AD/AS same as market demand/supply curve
  - We plot nominal variable (price level) against real variable (output)
  - Most economists use it to explain short-run fluctuations in economic activity
    - Around its long-run trend

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Short-Run Economic Fluctuations

- Aggregate-demand curve
  - Shows the quantity of goods and services that households, firms, the government, and customers abroad want to buy at each price level
  - Downward sloping?
  - AD = C + I + G + NX
  - Wealth effect, interest rate effect and exchange rate effect.

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Short-Run Economic Fluctuations

- Aggregate-supply curve
  - Shows the quantity of goods and services that firms choose to produce and sell at each price level
  - Long run AS curve however is vertical (classical assumptions hold)
  - Shifts in AD only increase inflation, and LRAS shifts over time due to increases in productivity and human capital.

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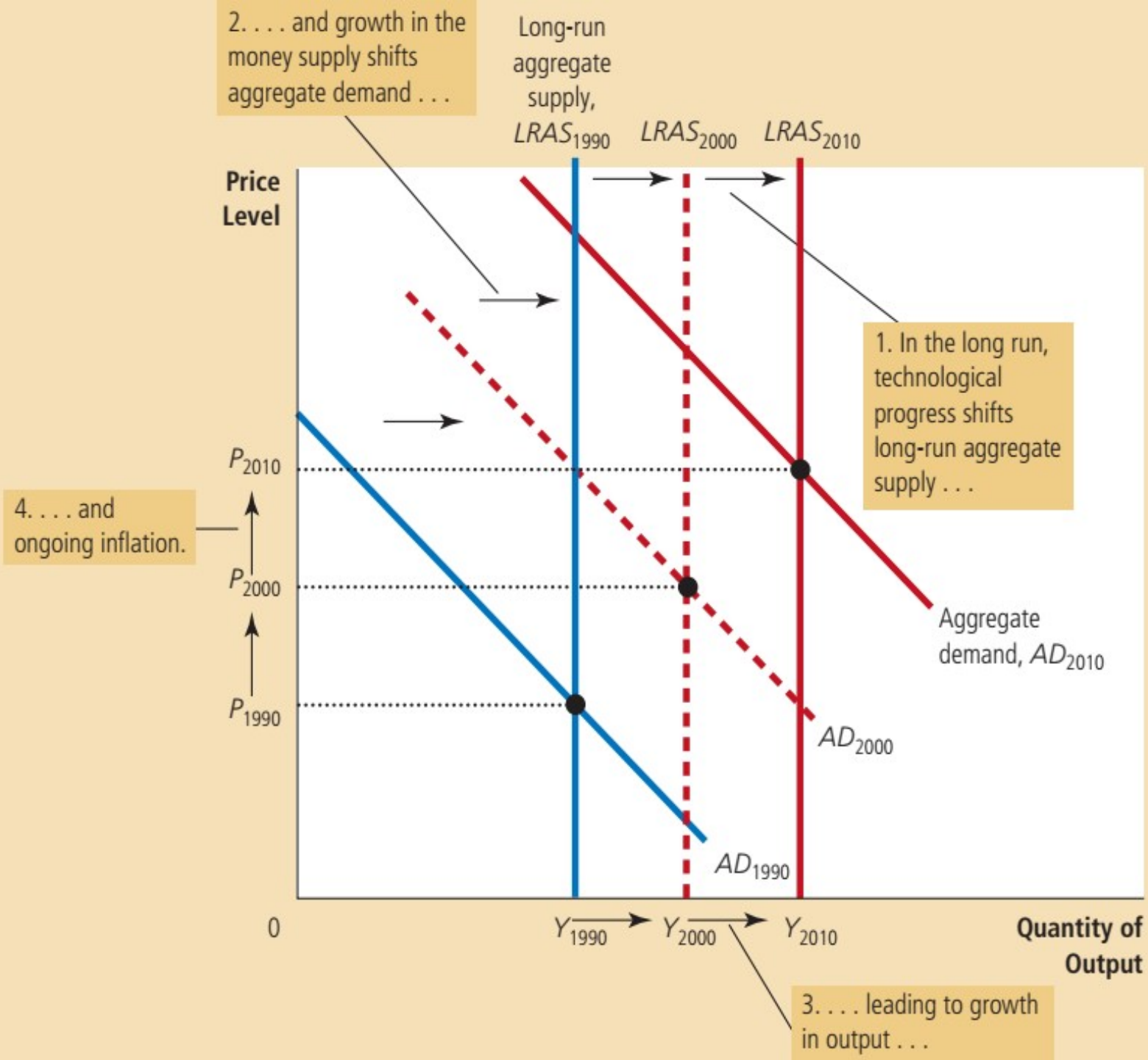
2. . . . and growth in the money supply shifts aggregate demand . . .
Long-run aggregate supply, $LRAS_{1990}$
$LRAS_{2000}$
$LRAS_{2010}$
Price Level
1. In the long run, technological progress shifts long-run aggregate supply . . .
P_{2010}
4. . . . and ongoing inflation.
P_{2000}
P_{1990}
Aggregate demand, AD_{2010}
AD_{2000}
AD_{1990}
0
Y_{1990}
Y_{2000}
Y_{2010}
Quantity of Output
3. . . . leading to growth in output . . .

Aggregate Supply Curve

- Short run
  - Aggregate-supply curve is upward sloping
  - A higher price raises the quantity of goods and services supplied

- Short run fluctuations are deviations from continuing long run trends of output growth and inflation.

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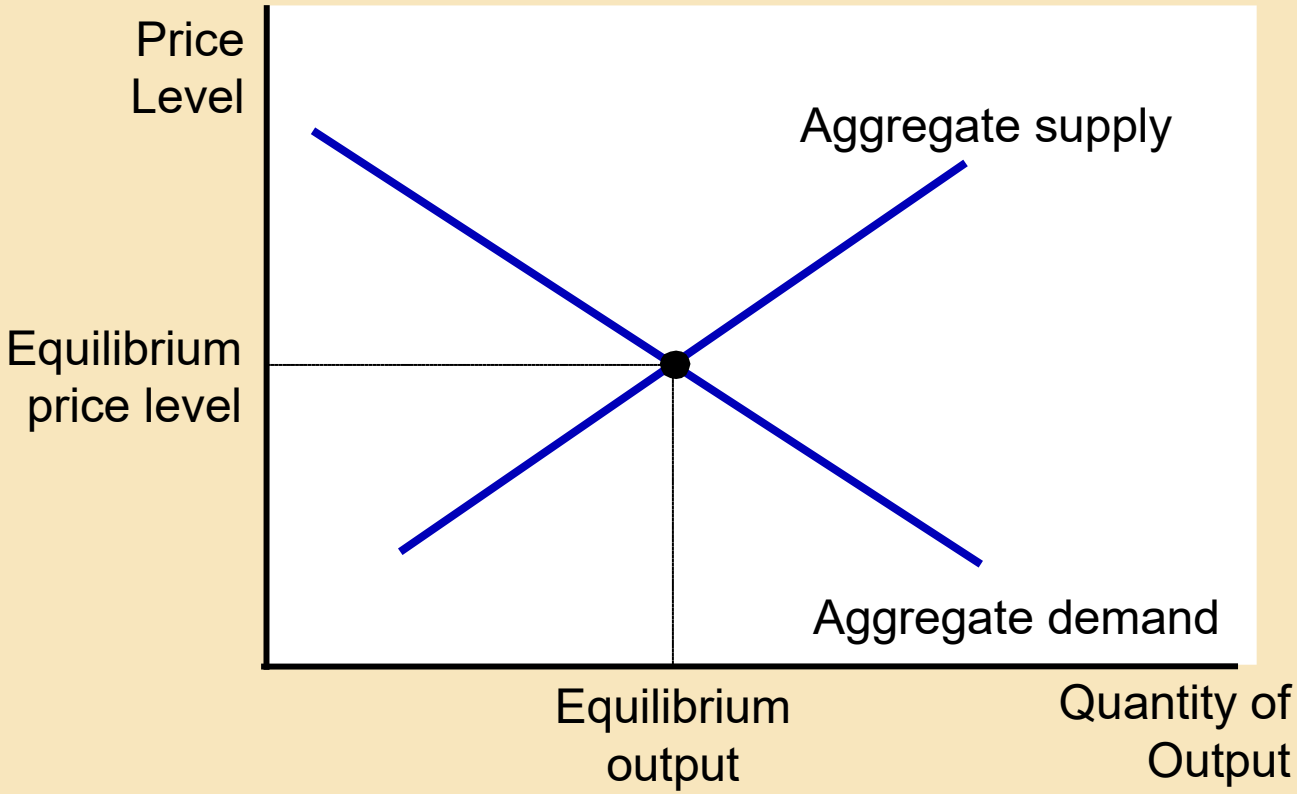
Figure 2

Aggregate Demand and Aggregate Supply



Economists use the model of aggregate demand and aggregate supply to analyze economic fluctuations. On the vertical axis is the overall level of prices. On the horizontal axis is the economy's total output of goods and services. Output and the price level adjust to the point at which the aggregate-supply and aggregate-demand curves intersect.

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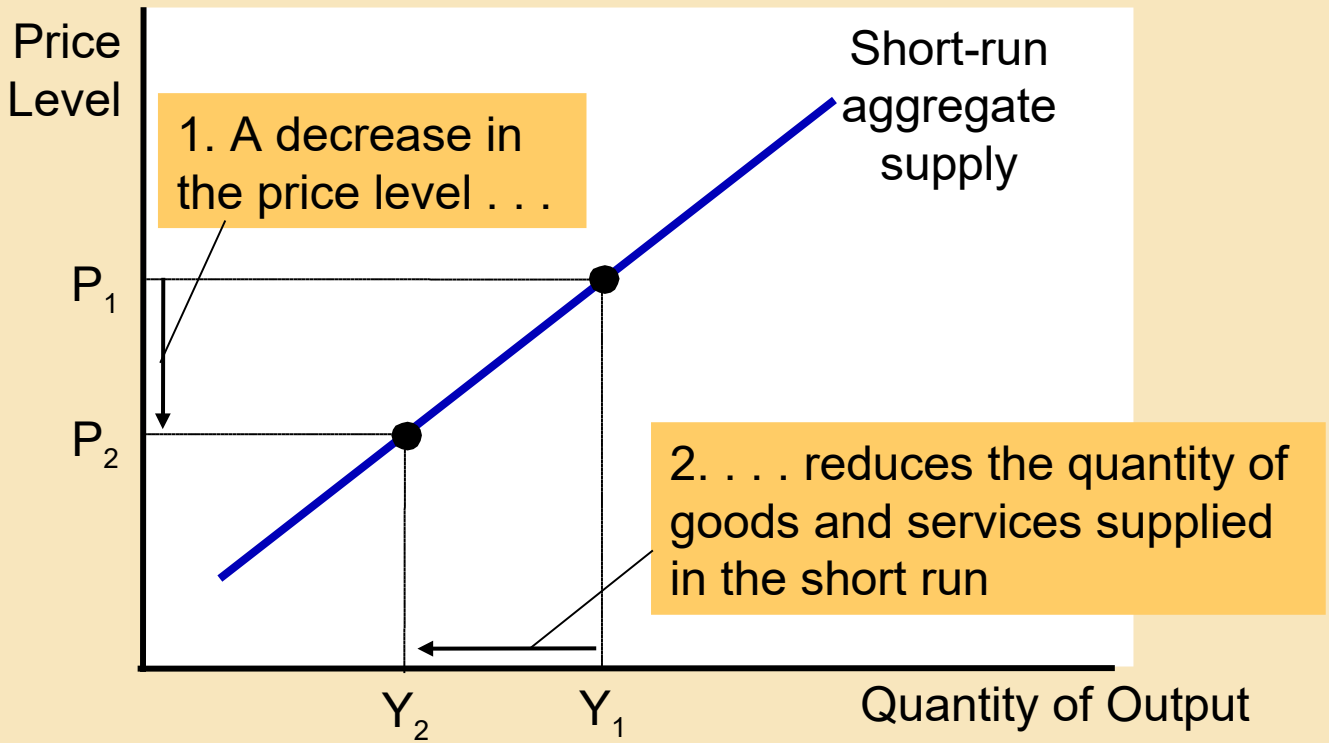
Figure 6

The Short-Run Aggregate-Supply Curve



In the short run, a fall in the price level from P_1 to P_2 reduces the quantity of output supplied from Y_1 to Y_2. This positive relationship could be due to sticky wages, sticky prices, or misperceptions. Over time, wages, prices, and perceptions adjust, so this positive relationship is only temporary.

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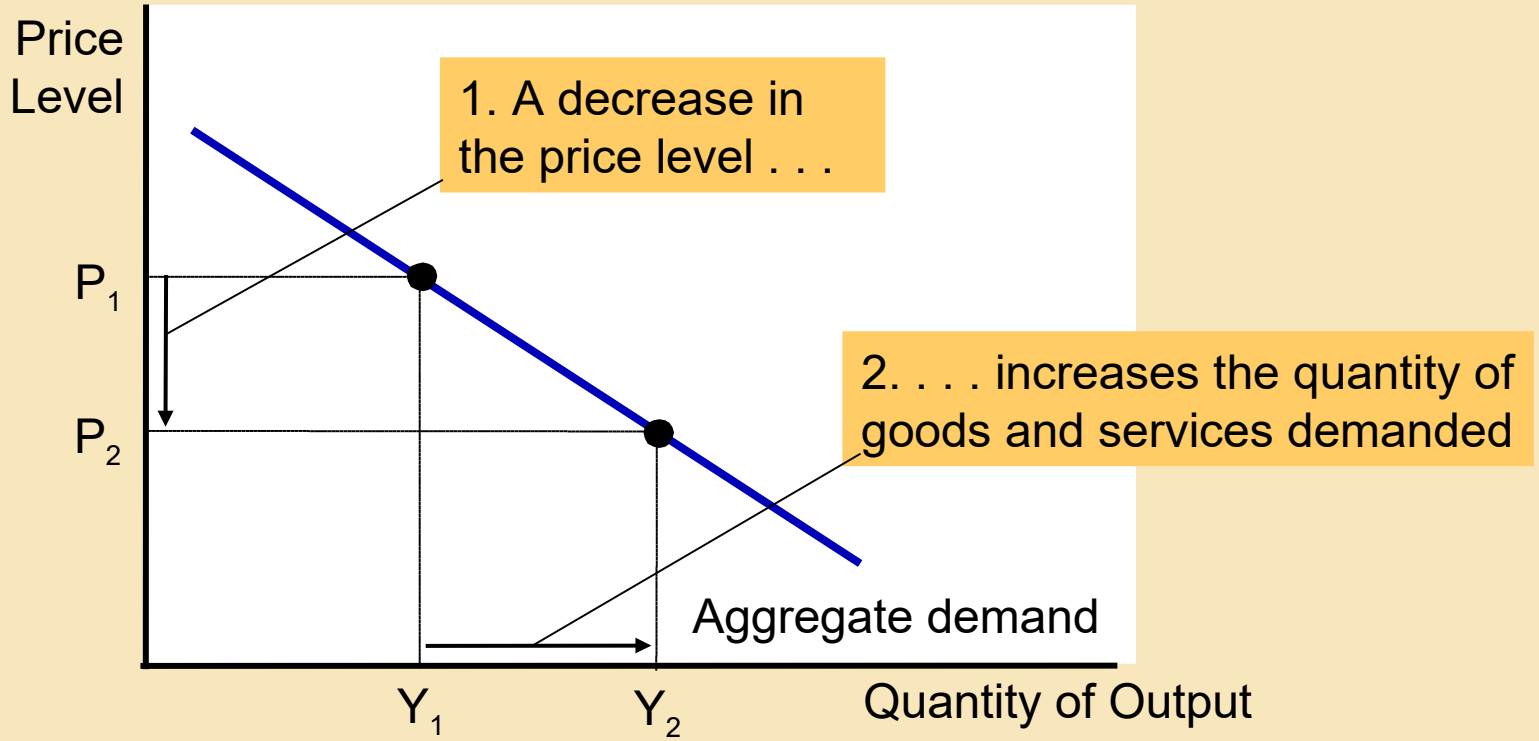
Figure 3

The Aggregate-Demand Curve



A fall in the price level from P_1 to P_2 increases the quantity of goods and services demanded from Y_1 to Y_2. There are three reasons for this negative relationship. As the price level falls, real wealth rises, interest rates fall, and the exchange rate depreciates. These effects stimulate spending on consumption, investment, and net exports. Increased spending on any or all of these components of output means a larger quantity of goods and services demanded.

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Upward sloping AS curve

- Why does AS slope upwards?
- Market imperfections/frictions/wedges cause supply to depend on price level
- Gap between expected price and actual price level causes long run supply to deviate from short run supply.
- *Sticky Wages*
- Quantity of output supplied = Natural rate of output + a(Actual price level – expected price level)

Aggregate-Demand Curve

- The AD curve might shift:
  - Changes in consumption
  - Changes in investment
  - Changes in government purchases
  - Changes in net exports

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Aggregate-Demand Curve

- Changes in government purchases, G
  - Policymakers – change government spending at a given price level
    - Build new roads
  - Increase in government purchases
    - Aggregate demand - shifts right

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Table 2

The Short-Run Aggregate-Supply Curve: Summary

Why Does the Short-Run Aggregate-Supply Curve Slope Upward?

1. *The Sticky-Wage Theory:* An unexpectedly low price level raises the real wage, which causes firms to hire fewer workers and produce a smaller quantity of goods and services.
2. *The Sticky-Price Theory:* An unexpectedly low price level leaves some firms with higher-than-desired prices, which depresses their sales and leads them to cut back production.
3. *The Misperceptions Theory:* An unexpectedly low price level leads some suppliers to think their relative prices have fallen, which induces a fall in production.

Why Might the Short-Run Aggregate-Supply Curve Shift?

1. *Shifts Arising from Changes in Labor:* An increase in the quantity of labor available (perhaps due to a fall in the natural rate of unemployment) shifts the aggregate-supply curve to the right. A decrease in the quantity of labor available (perhaps due to a rise in the natural rate of unemployment) shifts the aggregate-supply curve to the left.
2. *Shifts Arising from Changes in Capital:* An increase in physical or human capital shifts the aggregate-supply curve to the right. A decrease in physical or human capital shifts the aggregate-supply curve to the left.
3. *Shifts Arising from Changes in Natural Resources:* An increase in the availability of natural resources shifts the aggregate-supply curve to the right. A decrease in the availability of natural resources shifts the aggregate-supply curve to the left.
4. *Shifts Arising from Changes in Technology:* An advance in technological knowledge shifts the aggregate-supply curve to the right. A decrease in the available technology (perhaps due to government regulation) shifts the aggregate-supply curve to the left.
5. *Shifts Arising from Changes in the Expected Price Level:* A decrease in the expected price level shifts the short-run aggregate-supply curve to the right. An increase in the expected price level shifts the short-run aggregate-supply curve to the left.

Causes of Economic Fluctuations

- Assumption
  - Economy begins in long-run equilibrium
- Long-run equilibrium:
  - Intersection of AD and LRAS curves
    - Output - natural rate
    - Actual price level
  - Intersection of AD and short-run AS curve
    - Expected price level = Actual price level

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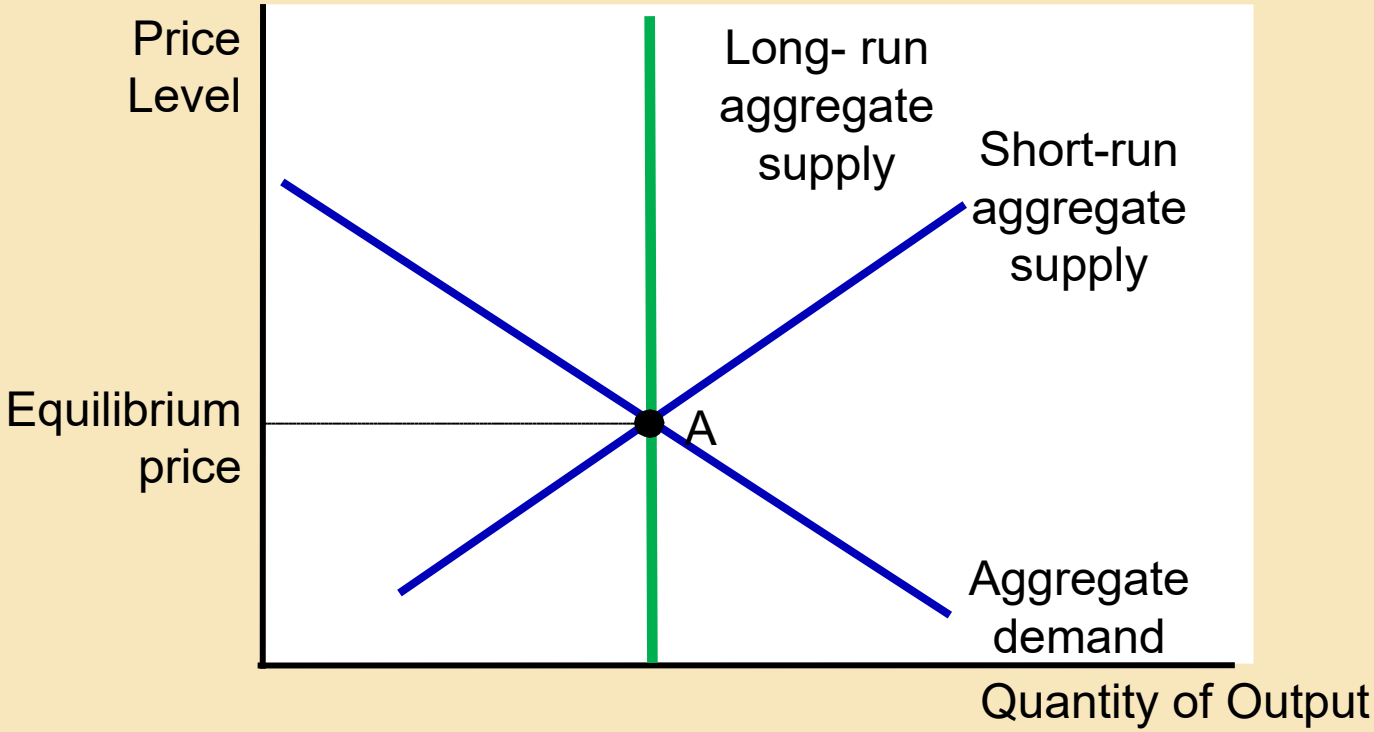
Exhibit 7

The Long-Run Equilibrium



The long-run equilibrium of the economy is found where the aggregate-demand curve crosses the aggregate-supply curve (point A).

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Causes of Economic Fluctuations

- Shift in aggregate demand
  - Wave of pessimism (stock market crash)
    - Aggregate demand shifts left
      - Short-run
      - Output falls
      - Price level falls

- Shift in aggregate supply

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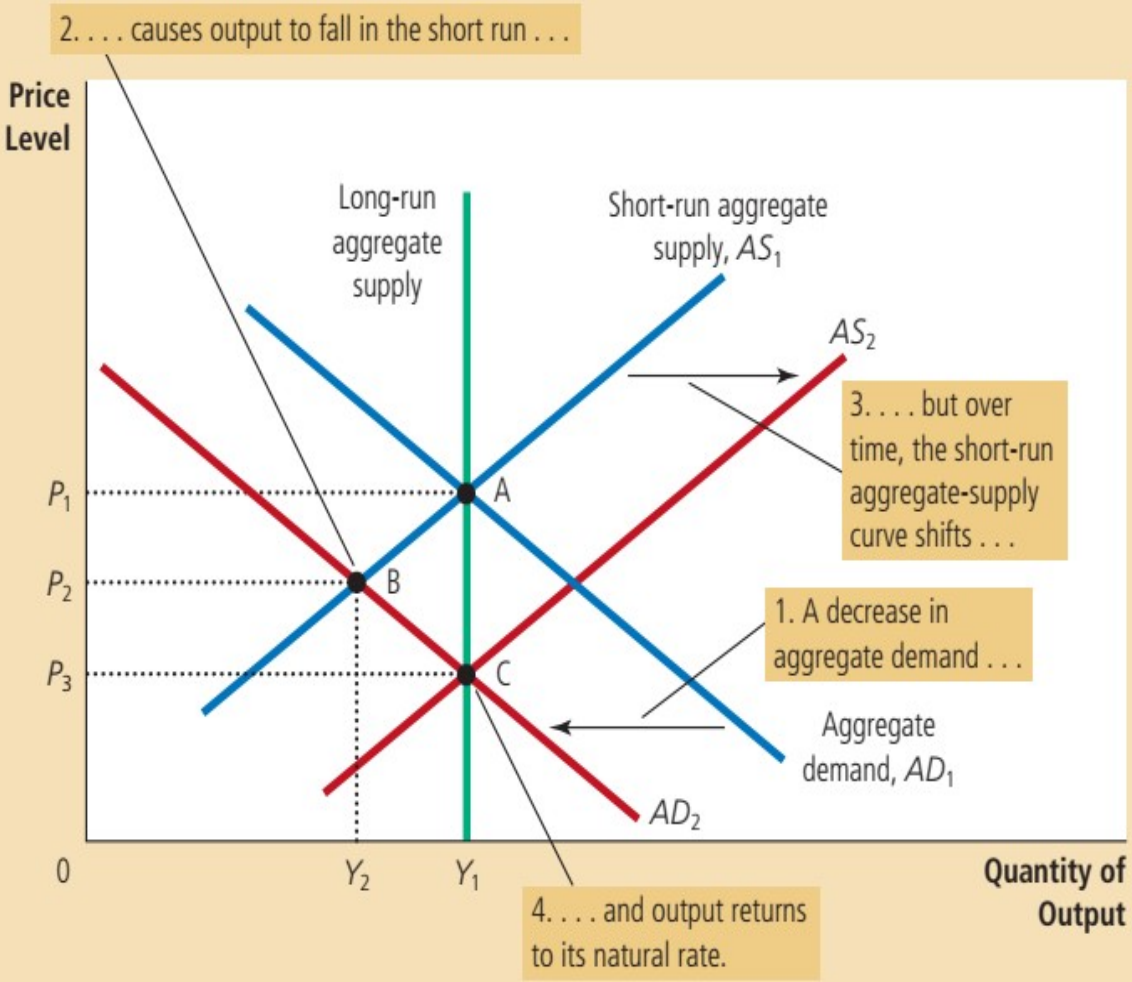
Figure 8

A Contraction in Aggregate Demand

A fall in aggregate demand is represented with a leftward shift in the aggregate-demand curve from AD_1 to AD_2. In the short run, the economy moves from point A to point B. Output falls from Y_1 to Y_2, and the price level falls from P_1 to P_2. Over time, as the expected price level adjusts, the short-run aggregate-supply curve shifts to the right from AS_1 to AS_2, and the economy reaches point C, where the new aggregate-demand curve crosses the long-run aggregate-supply curve. In the long run, the price level falls to P_3, and output returns to its natural rate, Y_1.



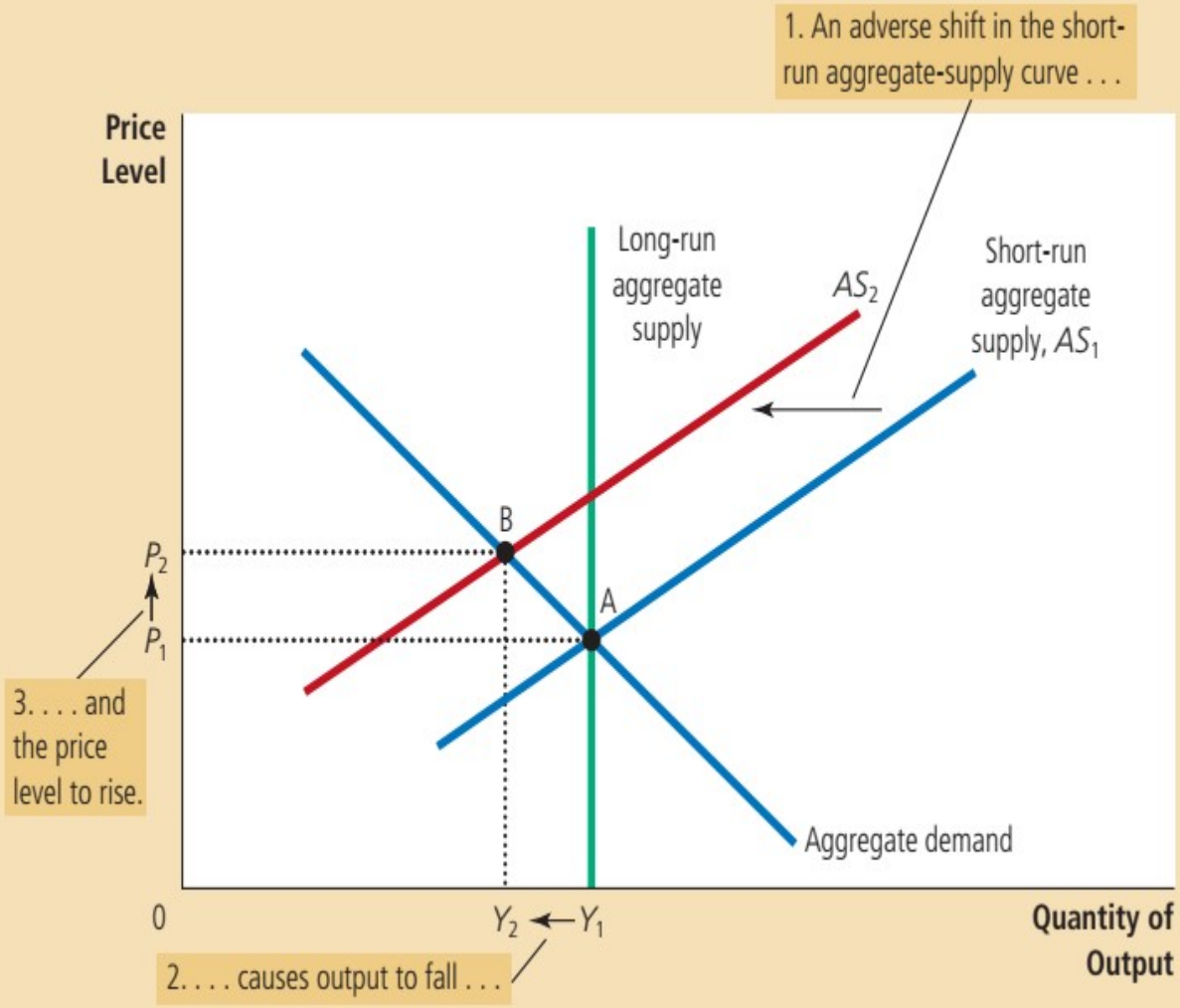


Figure 10

An Adverse Shift in Aggregate Supply

When some event increases firms' costs, the short-run aggregate-supply curve shifts to the left from AS_1 to AS_2. The economy moves from point A to point B. The result is stagflation: Output falls from Y_1 to Y_2, and the price level rises from P_1 to P_2.